Experiment 1

Aim

To design an ER diagram for an Indian e-commerce platform containing the entities Customer, Product, Order, OrderItem, Seller, Category, Payment, Delivery & Address and to represent primary keys, composite attributes, multivalued attributes, weak entities, specialization & participation constraints.

1. Entities & Attributes

Entity	Attributes
Customer	Customer_ID, Name, Email, Phone
Address	Address_ID, Address_Details
Product	Product_ID, Product_Name
Seller	Seller_ID, Seller_Name

2. Composite Attribute

→ For Customer, use:

Name
- First_Name
- Middle_Name
- Last_Name

→ For Address, use:

Address - Details
- House_No
- Street
- Area
- City
- State
- PIN_Code

3. Multi Valued Attribute

For Customer:

Phone

A customer may have more than one phone number.

Represent it using a double oval:

Customer → Phone

For the relational implement, a separate table named CustomerPhone will be created.

4. Weak Entity

Use OrderItem as the weak entity.

→ Identifying Relationship

Order — contains — OrderItem

→ OrderItem can be identified using:

Order_ID + Product_ID

Therefore:

- Order_ID → Foreign Key
- Product_ID → Foreign Key
- (Order_ID, Product_ID) → Primary Key

5. Specialization of Product

Specialize Product into:

- Electronics
- Clothing
- Grocery

Common attributes remains in Product.

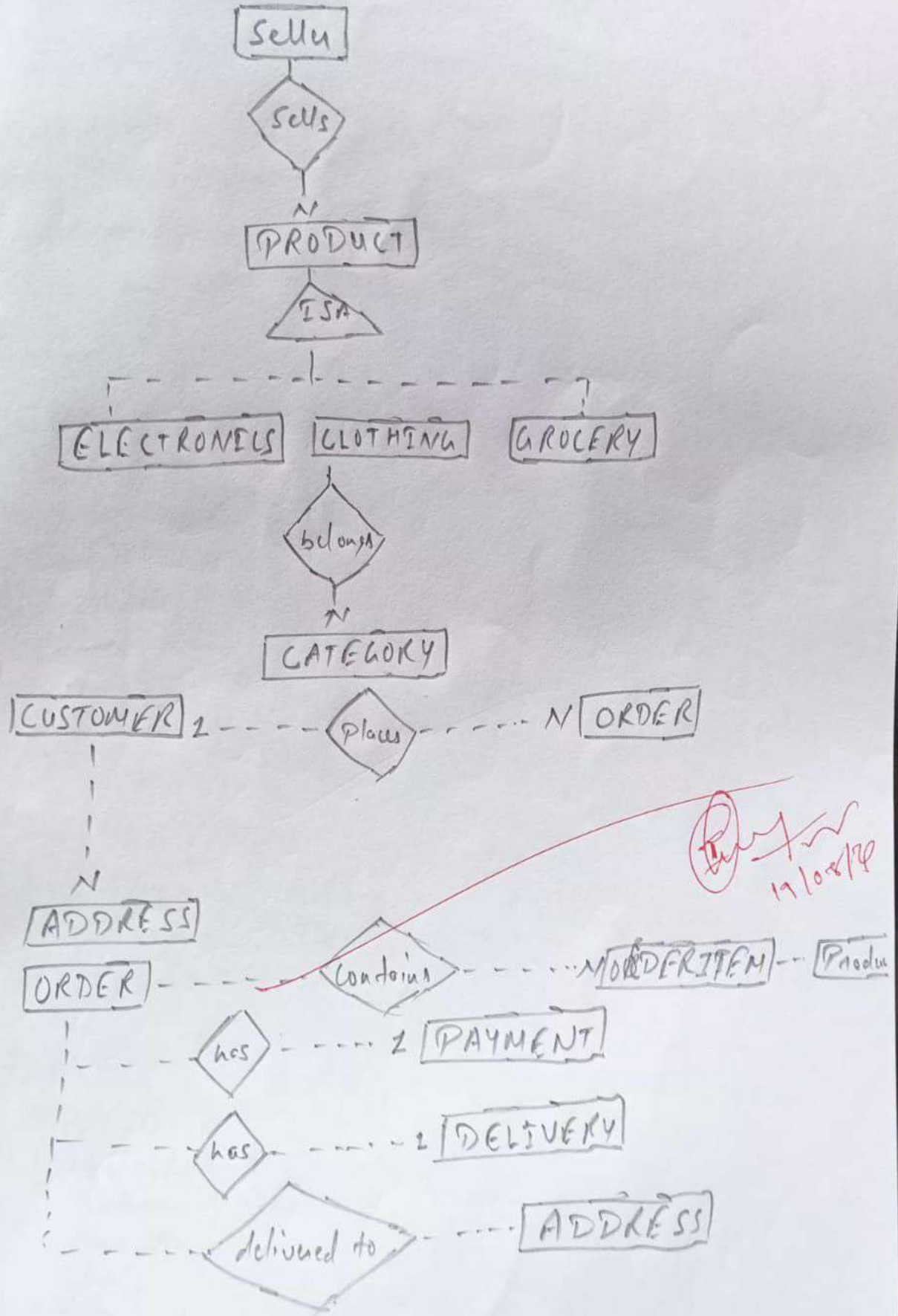
Electronics
- Warranty_Period

Clothing
- Size
- Fabric

Grocery
- Expiry_Date
- Weight

6. Relationship

Relationship	Cardinality	Participation
Customer – Places – order	1 : N	Order : Total, Customer : Partial
Customer – has – Address	1 : N	Address : " "
Seller – sells – Product	1 : N	Product : Total
Category – contains – Product	1 : N	"
Order – contains – OrderItem	1 : N	OrderItem : Total
Order – has – Payment	1 : 1	Payment : Total
Order – has – Delivery	1 : 1	Delivery : Total


Sellu
sells
N
PRODUCT
ISA
ELECTRONICS
CLOTHING
GROCERY
belongs
N
CATEGORY
CUSTOMER
1
places
N
ORDER
N
ADDRESS
ORDER
contains
M
ORDERITEM
Produ
has
1
PAYMENT
has
1
DELIVERY
delivered to
ADDRESS